

A Heuristic Solver for Rooted Maximum Agreement Forest

Aadit Nair, Anish Deodhar, Sumit Mishra
Contact: an25dsb0a35@student.nitw.ac.in

1. Representation, parsing, validation, and output

The solver is one native, mostly header-only C++ executable. `Tree` uses flat arrays for children, parent, depth, DFS entry/exit times, postorder, subtree leaf counts, sorted labels, dense label-to-leaf lookup, and a binary-lifting LCA table. Forests are sorted vectors of sorted label blocks rather than explicit fragments. Input is parsed from the PACE/Newick format and currently requires exactly two trees. Output Newick fragments are induced from the original first tree; singleton and two-label components have direct formatting paths.

For a label set C , validation builds the virtual induced tree from selected leaves and LCAs of adjacent leaves in DFS order. Unordered rooted child signatures are interned bottom-up and compared between T_1 and T_2 . The footprint $E_T(C)$ is the union of leaf-to-lca $_T(C)$ paths, representing each edge by its child endpoint. A forest is accepted only if no footprint edge is owned twice in either tree. The same topology and footprint routines are reused by packing, crossover, local repair, and component collapse.

2. Reductions, decomposition, and exact subproblems

`apply_reductions_exhaustively` performs up to 40 passes, subject to time and size gates, and records reversible entries in `ReductionStack`:

- Common-subtree reduction: equal canonical rooted-subtree signatures are contracted to a placeholder taxon. Expansion replaces the placeholder by all original labels in the same component.
- Common-chain reduction: a common rooted chain is truncated to three retained taxa. On expansion, the removed suffix is reattached when the retained prefix remains together; otherwise suffix labels are restored as singletons. This always preserves feasibility, but the fallback is not optimality-certifying.
- 3–2 chain reduction: in the implemented pendant 3-chain versus pendant 2-chain pattern, one recorded taxon is deleted and is always restored as a singleton. There is no separate objective “unit adjustment” in the output code.

Common clusters trigger recursive decomposition. A closed state unions independently solved bottom and top forests. When enabled by profile/size/depth/time gates, an open state solves marker-augmented bottom and top instances, merges the two marker-containing components after removing `X_DOWN` and `X_UP`, and validates the result. A hierarchy/template cache avoids rebuilding repeated cluster structures. Cluster bridge recovery is present as configuration machinery, but the baseline default is disabled; the active extended logic uses selectively gated open profiles and a late retry.

`TinyExactOracle` is used for very small whole or cluster instances (baseline defaults 10 and 12 leaves). It enumerates all nonempty topology-compatible subsets, stores their two edge masks, and runs a memoized branch-and-bound cover search branching on the first uncovered label, with incumbent, maximum-candidate-size, timer, and node-limit pruning. This is exact only for the submitted reduced subproblem.

3. Constructive and local heuristics

Warm start. `FastCherryApprox` matches canonical subtree signatures and greedily chooses larger node-disjoint common subtrees, then emits remaining labels as singletons. `CherryPairApprox` collects every cherry from each input tree (a two-label set is topology-compatible in either tree), sorts by total footprint size, and greedily packs label- and edge-disjoint pairs.

Active cherries. `ActiveCherryGreedyApprox` maintains an active first tree and a dynamic second-tree forest. Common cherries are contracted; otherwise sampled active cherries are scored and a cut/merge action is applied. Eleven policies are implemented: `Balanced`, `PreferSmallCuts`, `PreferBigProgress`, `PreferDifferentComponent`, `PreferLowConflictMass`, `PreferFewPendants`, `PreferImmediateGain`, `ConservativeSingleCut`, `DualityConservative`, `ResolveFinalCut`, and `AggressiveMultiCut`. Size-dependent portfolios vary policy, tree order, step multiplier, sample cap, local limit, deterministic salt, and shallow rank-choice scripts. Runs may periodically publish intermediate candidates, but all such candidates still pass the global validator.

Merge-only cleanup. `LocalImprove` performs bounded rounds of structural pair merges, singleton reinsertion, and ordinary pair merges. Each tentative component is checked for equal topology and for replaceable edge ownership in both trees. Exceptions return the original valid forest. It is enabled only below size gates (baseline 1,500 leaves, extended 2,500, with one late extended override to 5,000 in the main schedule).

4. Agreement-component packing

`AgreementComponentPacking` is the second main quality engine. A candidate stores its sorted labels, leaf ids, two edge footprints, gain $|C|-1$, edge cost, source flags, and deterministic tie hash. Candidate sources include components of seed forests, explicit/archive candidates, intersections of seed components, tree-order and pseudo-random pairs, small pair extensions, complete small subtrees from either tree, and neighbor-grown components. Components failing topology compatibility are rejected when inserted.

The restricted master starts from the best complete seed and runs several greedy candidate orders. It then applies selected-component exchange, bounded-deficit ejection/refill, and exact maximum-weight independent-set repair on small conflict neighborhoods. If enabled and the pool/leaf limits permit, a bounded branch-and-bound exact layer solves the generated candidate universe after forced and dominated-candidate processing. Uncovered labels are always added as singletons. Thus packing is exact only relative to its bounded candidate universe when that layer completes, not for the unrestricted rMAF instance.

5. Crossover, archives, and incumbent-focused generation

`ForestCrossover` combines two complete valid parent forests exactly over their union. After identical components are deduplicated, components inherited from the same parent cannot conflict; all remaining conflicts therefore cross the two parent sides. The maximum-weight independent set of this bipartite conflict graph, with weight $|C|-1$, is obtained as the complement of a minimum-weight vertex cover using one s-t min-cut. Labels not covered by the selected non-singleton components become singletons.

For $220 \leq n \leq 5,000$, `ComponentForestArchive` retains diverse validated forests and unique non-singleton components with support/use metadata. It supplies crossover parents and packing candidates. Main-level generators create additional strict candidates: nearby incumbent-component unions in both DFS orders; singleton rescue by insertion, two-singleton attachment, and label ejection; blocked-seed shattering; nearby seed unions; and direct component-collapse candidates consisting of small unions of incumbent components. Collapse is accepted only when the merged component is topology-compatible, its footprint conflicts solely with removed components, and the reconstructed forest validates.

Reduced-global diversification independently reapplies reductions and launches ResolveFinalCut, DualityConservative, or Balanced active-cherry variants, expands them, then uses direct comparison, crossover, and repacking. The extended booster varies salts, swap direction, and step budgets. For very large instances a Balanced seed bank feeds crossover, final-global packing, and large singleton-ejection repair. A small alternate exactification tail is implemented but explicitly disabled (`use\_small\_exactification\_tail=false`) in the submitted active schedule.

6. Correctness scope and provenance

Feasibility follows from the incumbent invariant: the singleton forest is valid; every published replacement partitions X , has equal induced topology in both trees, and has disjoint induced edges; and all reduced candidates are expanded and rechecked on the original instance. Exactness claims are local: the tiny oracle is exact for its bounded subproblem; crossover is exact over the two parent component sets; local MWIS and bounded packing search are exact only on their selected conflict pools/candidate universe. The solver makes no global optimality claim.

The common-subtree/common-chain preprocessing, cluster reduction, and active-cherry cases are rooted-MAF/rSPR ideas from prior work [1–6]. The implemented 3–2 rule follows the strengthened rooted kernel literature [5]. The component formulation is related to ILP/LP and duality-based MAF formulations [7,8], while this solver's restricted candidate generation, archives, policy portfolio, exact bipartite crossover, incumbent-collapse and singleton-repair generators, emergency publication, and two-deadline scheduling are implementation-specific combinations rather than direct reproductions of a cited algorithm.

References

- [1] M. Bordewich and C. Semple. On the computational complexity of the rooted subtree prune and regraft distance. *Annals of Combinatorics* 8(4):409–423, 2005. doi:10.1007/s00026-004-0229-z.
- [2] M. Bordewich, C. McCartin, and C. Semple. A 3-approximation algorithm for the subtree distance between phylogenies. *Journal of Discrete Algorithms* 6(3):458–471, 2008. doi:10.1016/j.jda.2007.10.002.
- [3] C. Whidden, R. G. Beiko, and N. Zeh. Fixed-Parameter Algorithms for Maximum Agreement Forests. *SIAM Journal on Computing* 42(4):1431–1466, 2013. doi:10.1137/110845045.
- [4] S. Linz and C. Semple. A Cluster Reduction for Computing the Subtree Distance Between Phylogenies. *Annals of Combinatorics* 15:465–484, 2011. doi:10.1007/s00026-011-0108-3.

- [5] S. Kelk, S. Linz, and R. Meuwese. Cyclic generators and an improved linear kernel for the rooted subtree prune and regraft distance. *Information Processing Letters* 180:106336, 2023. doi:10.1016/j.ipl.2022.106336.
- [6] Z. Li and N. Zeh. Computing Maximum Agreement Forests without Cluster Partitioning Is Folly. *ESA 2017, LIPIcs* 87, 56:1–56:14. doi:10.4230/LIPIcs.ESA.2017.56.
- [7] Y. Wu. A practical method for exact computation of subtree prune and regraft distance. *Bioinformatics* 25(2):190–196, 2009. doi:10.1093/bioinformatics/btn606.
- [8] N. Olver, F. Schalekamp, S. van der Ster, L. Stougie, and A. van Zuylen. A duality based 2-approximation algorithm for maximum agreement forest. *Mathematical Programming* 198:811–853, 2023. doi:10.1007/s10107-022-01790-y.